Chapter 1: Organization Structure of Industry and General Layout

Introduction

SoftTech Computer Training Institute is a leading IT training organization dedicated to bridging the gap between academic learning and industry requirements. It was established with the vision of providing quality technical education and enhancing the employability of students through industry-oriented training and practical learning. The institute focuses on developing technical, analytical, and professional skills by offering training in emerging technologies such as Python Programming, Data Science, Data Analytics, Microsoft Power BI, Power Query, DAX, Machine Learning, Web Development, and Generative AI.

The institute works in collaboration with educational institutions and industry professionals to conduct certification programs, internships, workshops, live projects, and skill development initiatives. Through experienced trainers, modern infrastructure, and hands-on project-based learning, SoftTech provides students with practical exposure and technical expertise aligned with current industry standards, preparing them for successful careers in the Information Technology and Data Analytics domains.

Organization Structure

Soft Tech Innovations follows an organizational structure that supports both technical training and software-related activities. The organization is managed by the management team, which oversees the overall functioning of the institute and its different departments.

The major functional areas of the organization include Management, Training & Development, Software Development, and Data Science & Data Analytics. Each department has its own responsibilities and contributes to the overall working of the organization.

Major Functional Areas Management

The management is responsible for the overall administration and functioning of the organization. It handles planning, coordination, infrastructure, training activities, and other organizational requirements.

Training & Development

The Training & Development department focuses on providing technical education and practical training to students. The department conducts courses and training programs in different areas of software and technology.

Software Development

The Software Development department is concerned with software-related activities and development work. It provides an environment where students and trainees can gain exposure to practical software development concepts and technologies.

Data Science & Data Analytics

The Data Science and Data Analytics department focuses on areas such as data analysis, Python programming, data visualization, Power BI, and related technologies. The industrial

training program described in this report was conducted under this department.
The interns received training and practical guidance in Data Science and Data Analysis using Python and Power BI. They also worked on projects involving Exploratory Data Analysis and Power BI report development.

1.1 General Layout of the Organization

Soft Tech Innovations operates training centres in Kolhapur and Sangli.
The Kolhapur centre consists of three classrooms and one long computer laboratory. The classrooms are primarily used for training sessions, lectures, demonstrations, and practical activities. The laboratory provides computer-based working space when required.
The Sangli centre, which was inaugurated in 2024, consists of two classrooms and a smaller laboratory. The available infrastructure is arranged to support classroom-based technical training as well as practical sessions.
During the industrial training period, most interns worked using their own laptops at their respective desks. Because of this arrangement, the computer laboratory was not used extensively for the internship activities. The interns were generally able to perform programming, data analysis, Power BI development, and project-related activities using their personal systems.
The training environment provided a suitable workspace for individual as well as guided learning. Trainers could conduct demonstrations and explain concepts in the classroom, while interns could simultaneously work on exercises and project tasks using their laptops.

1.2 Working Environment

The organization provides a learning-oriented environment where students and interns can interact with trainers while working on technical activities. The internship structure combines theoretical explanation with practical implementation.
For the Data Science and Data Analytics internship, the working environment was mainly focused on learning Python for data analysis, performing Exploratory Data Analysis, creating visualizations, developing Power BI reports, and understanding the use of Generative AI.
This structure allowed the interns to gradually move from basic concepts to practical tasks and project implementation during the 12-week industrial training period.

Chapter 2: Introduction to Industry / Organization

2.1 Introduction

Soft Tech Innovations, formerly known as Soft Tech Computer Training Institute, is a software training and technology services organization established in 2011. The organization is based in Kolhapur, Maharashtra, and expanded its presence with a new branch in Sangli in 2024.

With more than a decade of experience in the technology and training field, the organization focuses on providing job-oriented training, practical technical education, and real-world project applications. Along with training activities, Soft Tech Innovations also provides software development and software-related services to both domestic and international clients.

The organization aims to create a connection between academic learning and the practical requirements of the technology industry. Instead of limiting learning to theoretical concepts, importance is given to hands-on practice, practical assignments, project development, and real-world examples.

The organization has developed its training approach based on the understanding that students need more than academic knowledge to enter the professional environment. They also need practical skills, problem-solving ability, confidence, communication skills, and an understanding of how technologies are used in actual projects.

2.2 Background of the Organization

Soft Tech Computer Training Institute was established in 2011 with the objective of providing quality technical education and helping students develop skills required for employment in the software industry.

Over the years, the organization has worked with students from different educational backgrounds and has continued to expand its technical training activities. With changing requirements in the software industry, the organization has also adapted its training approach to include new technologies, practical applications, project-based learning, and industry-oriented concepts.

The organization later adopted the name Soft Tech Innovations, reflecting its broader involvement in both technology training and software development and services.

The organization currently operates from its centres in Kolhapur and Sangli, providing an environment for technical learning, practical work, project development, and professional guidance.

2.3 Nature of the Organization

Soft Tech Innovations operates in two closely connected areas: technology education and software services.

The training side of the organization focuses mainly on job-oriented training. The objective is to help learners understand technical concepts and apply them practically. Training generally follows a combination of theory and practical implementation, where concepts are first explained and then applied through exercises, assignments, and projects.

The organization also undertakes software development and related services for domestic as well as international clients. This provides exposure to practical software requirements and helps maintain an industry-oriented approach toward technical training.

The combination of training and software development allows the organization to maintain a practical perspective when designing learning activities and projects.

2.1 Target Learners and Training Activities

Soft Tech Innovations works with learners from different educational and professional backgrounds.

The organization mainly provides training and technical learning opportunities to:

- College students
- Fresh graduates
- School students
- Corporate employees
- Faculty members through Faculty Development Programs
- Students participating in short-term workshops

The organization also conducts 6-day workshops for school and college students, where participants are introduced to practical concepts and technologies within a limited period.

The training environment is designed to accommodate learners with different levels of technical knowledge. Greater importance is given to understanding concepts through practical implementation rather than only completing theoretical portions.

2.2 Training Approach

The general training methodology followed by Soft Tech Innovations is based on a combination of theory and practical learning.

Theoretical concepts are introduced first so that students understand the purpose and working of a particular technology or concept. These concepts are then followed by practical exercises that allow students to implement what they have learned.

The major components of the training approach include:

Theory and Practical Learning

Technical concepts are explained before students begin implementation. This helps learners understand not only *how* something works but also *why* it is used.

Hands-on Practice

Students are encouraged to work directly with the technologies being taught. Practical implementation helps them gain confidence and become familiar with the tools used in technical work.

Assignments

Assignments are used to allow students to practice concepts independently. They also help trainers understand whether students are able to apply the concepts after the training session.

Real-World Examples

Where appropriate, concepts are explained using examples related to practical or real-world applications. This makes it easier for students to understand how technical knowledge can be applied outside the classroom.

Project-Based Learning

Projects form an important part of the practical learning process. Working on projects allows students to combine multiple concepts and develop a complete solution instead of working only on individual programming exercises.

2.1 Industrial Training Program

As part of the industrial training program, students undertook 12 weeks of training in the domain of Data Science and Data Analytics at Soft Tech Innovations.

The main objective of this internship was to provide students with practical exposure to Data Science and Data Analytics and help them apply their theoretical knowledge to real-world datasets and reporting requirements.

The internship included practical work using Python and Power BI, along with project development and other professional development activities.

The internship was structured around two major projects.

Project 1: Exploratory Data Analysis Using Python

The first project focused on Exploratory Data Analysis (EDA) using Python

Students worked with datasets and learned how to examine, clean, analyze, and understand data before drawing conclusions from it. The project provided practical exposure to the process of working with datasets and presenting useful information through analysis and visualization. The project helped students understand how Python can be used as a tool for data cleaning, analysis, visualization, and interpretation.

Project 2: Power BI Multipage Report and Analysis

The second project focused on Power BI and interactive report development.

Students worked on creating a multipage Power BI report, where different pages were designed to present data from different perspectives. The project involved working with data, creating visualizations, organizing report pages, and presenting analytical findings in an understandable format.

This project provided practical experience in transforming analyzed data into an interactive and visually understandable business report.

2.1 Generative AI Introduction

The internship also included an introduction to Generative Artificial Intelligence (GenAI).

The purpose of this component was to make students aware of the growing role of AI in the technology industry and introduce them to the basic concepts and practical applications of Generative AI.

The introduction was included as an additional learning component rather than as the primary focus of the internship. It helped students understand how modern AI tools can support activities such as learning, productivity, content generation, analysis, and problem-solving.

2.2 Professional Grooming and Career Preparation

The final stage of the internship also included activities related to professional grooming and

career preparation.
Students were provided guidance on areas that are important when entering the professional environment. One of the major activities was resume building, where students were guided on how to organize their educational background, technical skills, projects, and other relevant information in a professional manner.
Professional grooming sessions were also conducted to help students improve their overall readiness for employment.

when applying for jobs. Students also need to be able to present their skills, explain their projects, communicate effectively, and approach professional opportunities with confidence.

2.1 Internship Monitoring

Unlike a regular academic course, the industrial training program did not follow a formal examination or assessment system.
The students were mainly evaluated through their practical work and assignments. Assignments were given during the training period and their progress was monitored by the trainers.
The focus was on whether students could understand the concepts, implement them practically, complete assigned work, solve problems, and gradually work independently.
This approach provided students with an environment where they could learn through implementation rather than focusing primarily on marks or examinations.

2.2 Vision of the Organization

The vision of Soft Tech Innovations can be described as:

To provide practical, industry-oriented and job-focused technical education while helping learners develop the skills, confidence and professional knowledge required to build successful careers in the technology industry.

The organization aims to continue developing an environment where students can learn current technologies, work on practical projects, and understand how their knowledge can be applied in real-world situations.

2.3 Strengths of the Organization

Some of the important strengths of Soft Tech Innovations include:

Experienced Trainers

The organization has experienced trainers who guide students through technical concepts and practical implementation. Their experience helps students understand concepts from both academic and practical perspectives.

Practical Training

These activities were particularly useful because technical knowledge alone is not sufficient

Greater importance is given to practical learning and hands-on implementation, allowing students to gain experience working with technologies rather than only studying their theoretical concepts.

Industry-Oriented Curriculum

The training content is designed with consideration for industry requirements and current technology practices, helping students develop skills that are relevant to professional opportunities.

Project-Based Learning

Students get opportunities to work on projects and practical applications, allowing them to combine multiple concepts and develop problem-solving skills.

Long Experience

Since its establishment in 2011, the organization has gained considerable experience in the field of technical education and training.

Student Support

Students receive guidance during their learning process, including support with technical doubts, assignments, practical work, and project-related activities.

Career Guidance

The organization also provides career-oriented guidance, including professional grooming and resume-building activities, particularly for students preparing to enter the job market.

Alumni Reviews

Feedback and reviews from alumni also contribute to the organization's reputation and provide an indication of the learning experiences of students who have previously completed their training.

2.4 Relevance of the Organization to the Internship

Soft Tech Innovations provided a suitable environment for the industrial training because the organization combines technical education, practical training, project development, and software services.

The internship allowed students to move beyond classroom-based learning and experience a structured working environment where they were required to apply their knowledge to datasets, analytical tasks, reports, and projects.

The 12-week training program therefore served as a bridge between academic knowledge and practical application. Through Python-based Data Science work, Power BI reporting

project assignments, Generative AI introduction, and professional grooming activities, students gained exposure to both technical and professional aspects of the technology industry.

Overall, the organization provided an environment where interns could gradually develop their technical knowledge, practical skills, analytical thinking, project experience, and confidence for entering the professional world.

Chapter 3: Major Hardware/Software Used

3.1 Introduction

During the 12-week industrial training program, the interns used a combination of hardware and software tools for learning, practical implementation, data analysis, project development, documentation, and presentation.

Since most interns worked on their own laptops, the hardware requirements were kept practical and suitable for the software used during the training. The main technical work involved Python programming, Data Science and Exploratory Data Analysis (EDA), Power BI report development, Git/GitHub, and Generative AI tools.

The software environment was selected in such a way that students could gradually move from basic programming to practical data analysis and finally to report development and presentation.

3.2 Hardware Used

The primary hardware used during the internship was the personal laptop of each intern. Since students worked individually on their systems, the exact configuration varied from student to student. However, the systems were required to meet reasonable specifications for running Python environments, Google Colab through a web browser, Power BI Desktop, and other supporting applications.

General Hardware Requirements

Hardware Component	General Requirement / Configuration	Purpose
Laptop/Desktop	Modern laptop or desktop computer	Main system for training and project work
Processor	Intel Core i3/i5 or equivalent AMD processor and above	Running Python, Power BI and other applications
RAM	Minimum 8 GB recommended	Handling datasets, Power BI reports and multiple applications
Storage	SSD preferred, with adequate free space	Faster system and application performance
Display	Standard laptop/monitor display	Coding, analysis and report development
Keyboard & Mouse	Standard keyboard and mouse/touchpad	Programming and general computer operation
Internet Connection	Stable internet connection	Google Colab, downloading datasets, documentation, Git/GitHub and AI tools
Screen/Display System	Screen display used during demonstrations/presentations	Training sessions and project presentations

The above specifications represent general requirements rather than a fixed configuration, since interns used their own systems.

3.1 Operating System

The interns primarily used the Windows operating system, with both Windows 10 and

Windows 11 systems being used during the training.
Windows provided compatibility with the major applications used in the internship, particularly Power BI Desktop, PyCharm, web browsers, Git/GitHub tools, and presentation software.

3.2 Software and Development Tools

3.2.1 PyCharm IDE
PyCharm was used during the initial Python programming sessions.
The IDE provided students with an environment where they could write, execute, and debug Python programs. It was mainly used while learning basic Python programming concepts before moving toward Data Science and EDA.
Students practiced programming concepts such as variables, data types, operators, conditional statements, loops, functions, collections, and other fundamental Python concepts.

3.2.2 Google Colab

Google Colab was primarily used for the Data Science and Exploratory Data Analysis (EDA) part of the internship.
It provided a convenient notebook-based environment where students could write Python code and execute it in separate cells. Since Colab works through a web browser, students did not need to maintain a complex local Data Science environment on their laptops.
Google Colab was particularly useful for:

- Data loading and exploration
- Data cleaning
- Data preprocessing
- Exploratory Data Analysis
- Data visualization
- Feature-related operations
- Scaling and preprocessing
- Working with Python Data Science libraries

Students also received an introduction to using Gemini AI within the Google Colab environment as part of the Generative AI component of the internship.

3.3 Python Libraries Used

Python formed an important part of the Data Science and EDA project. Several commonly used Python libraries were introduced and used for practical analysis.

Pandas

Pandas was used mainly for working with structured data.

Students used Pandas for activities such as:

- Reading CSV and Excel files
- Creating and manipulating DataFrames
- Examining datasets
- Selecting and filtering data
- Handling missing values
- Data cleaning
- Data transformation
- Basic statistical analysis

NumPy

NumPy was introduced for numerical operations and working with numerical data.

It supported activities involving:

- Arrays
- Numerical calculations
- Mathematical operations
- Data manipulation
- Numerical preprocessing

Matplotlib

Matplotlib was used for creating basic data visualizations.
Students learned how to represent data graphically using charts such as:

- Line charts
- Bar charts
- Histograms
- Scatter plots
- Other basic visualizations

Seaborn

Seaborn was used for creating more informative and visually detailed statistical visualizations. It was particularly useful during the EDA project, where students needed to identify patterns, relationships, distributions, and trends within datasets.

Scaling and Preprocessing Libraries

Additional Python libraries and functions were introduced where required for data preprocessing and feature scaling.
Scaling techniques were discussed to help students understand how numerical features can be brought to comparable ranges before being used in further analytical or machine learning-related operations.

3.4 Microsoft Power BI Desktop

Microsoft Power BI Desktop was one of the primary tools used during the second internship project.
Students used Power BI Desktop to transform analyzed data into interactive and visually

understandable reports.

The Power BI project focused mainly on creating a multipage report. Students worked with data, selected appropriate visualizations, organized report pages, and presented analytical information in a structured manner.

The major activities included:

- Importing data
- Working with CSV and Excel files
- Understanding data fields
- Creating visualizations
- Designing report pages

- Creating a multipage report
- Adding interactive elements
- Applying filters and slicers where required
- Organizing report layouts
- Presenting analytical findings

Students were also given an introduction to Power BI Service to understand how Power BI reports can be used in an online environment. However, the primary development work during the internship was performed using Power BI Desktop.

3.5 Microsoft PowerPoint

Microsoft PowerPoint was used for documentation and presentation-related activities.

At the end of the project work, students prepared presentations explaining their individual projects. Students presented their project work using PPTX presentations, along with screen recordings/demonstrations of their work.

This activity helped students practice:

- Project explanation
- Presentation skills
- Technical communication
- Explaining methodology and results
- Demonstrating project output
- Professional presentation

The presentation activity was also useful for developing confidence in explaining technical work to others.

3.6 Git and GitHub

Git and GitHub were introduced as tools for version control and project management.

Students were given an understanding of how project files can be managed using version control systems and how repositories can be maintained using GitHub.

The introduction helped students understand concepts such as:

- Repository
- Commit
- Push

- Pull
- Version history
- Project repository
- Sharing project work through GitHub

GitHub also provided students with an opportunity to maintain their project work in an organized manner.

3.7 Generative AI Tools

Generative AI tools were included as part of the internship to introduce students to the role of AI in modern technical and professional work.

Gemini AI

Gemini AI was introduced in connection with Google Colab. Students were shown how Generative AI can assist with understanding code, exploring ideas, and supporting Data Science-related learning.

The purpose was not to replace the student's own work but to demonstrate how AI tools can be used as an additional learning and productivity resource.

ChatGPT

ChatGPT was used mainly as a supporting tool for concept understanding and learning assistance.
Students could use it to clarify programming concepts, understand errors, obtain explanations of unfamiliar concepts, and explore alternative approaches to technical problems.
The emphasis was placed on understanding the concepts rather than directly depending on AI-generated answers.

3.8 Data File Formats Used

The internship projects primarily involved CSV and Excel files as data sources.

CSV Files

CSV files were used extensively during the Python-based Data Science and EDA activities. Students learned how to load CSV datasets into Python and perform various data analysis and visualization operations.

Excel Files

Excel files were also used as a source of structured data, particularly while working with Power BI.
Students learned how data stored in Excel could be imported into Power BI and converted into useful visual reports.

3.9 Internet and Online Resources

A stable internet connection was an important requirement throughout the internship.

Internet access was required for:

- Using Google Colab
- Accessing online documentation
- Downloading datasets
- Installing or updating required software and libraries
- Working with GitHub
- Accessing Power BI-related resources
- Using Gemini AI
- Using ChatGPT
- Researching technical concepts
- Accessing learning resources and references

The use of online resources also helped students develop the habit of finding technical information and referring to documentation when required.

3.10 Overall Software Environment

The major software and tools used during the internship can be summarized as follows:

Software / Tool	Main Purpose
Windows 10 / Windows 11	Operating system
PyCharm IDE	Basic Python programming
Google Colab	Data Science and EDA
Python	Programming and Data Analysis
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization and EDA
Scaling / Preprocessing Libraries	Data preprocessing and feature scaling
Power BI Desktop	Data analysis and multipage report development
Power BI Service	Introduction to online Power BI environment
Microsoft PowerPoint	Project presentation
Git	Version control
GitHub	Repository and project management
Gemini AI	Generative AI introduction and learning assistance
ChatGPT	Concept understanding and technical assistance
Web Browser	Accessing Colab, documentation and online resources

3.11 Role of Hardware and Software in the Internship

The hardware and software used during the internship were selected according to the practical requirements of the training. Since students worked primarily on their own laptops, they were able to continue their practical work outside the training centre as well.

The combination of PyCharm, Google Colab, Python Data Science libraries, Power BI Desktop, Git/GitHub, and Generative AI tools provided students with exposure to different stages of a data-oriented workflow.

The tools also helped students move progressively from basic programming → Data Science and EDA → data visualization → Power BI reporting → project presentation → professional and AI-assisted learning.

Overall, the software environment provided during the internship was sufficient for students to complete their assigned work and develop practical familiarity with the tools commonly associated with Data Science and Data Analytics.

Chapter 4: Processes / Methodologies

4.1 Introduction

The 12-week industrial training program at Soft Tech Innovations was designed to provide interns with a gradual transition from basic technical concepts to practical project development.

The training methodology focused on theory, hands-on implementation, assignments, problem-solving, and project-based learning.
The internship was conducted in a structured sequence so that students could first develop the required programming foundation and then apply that knowledge to Data Science, Exploratory Data Analysis, Power BI, and project development.
The overall learning progression followed:
Basic Python → Advanced Python and Data Handling → Data Science → Exploratory Data Analysis Project → Power BI → Multipage Power BI Project → Generative AI Introduction → Professional Grooming and Resume Building
The methodology was designed to make students increasingly independent in their technical work. Rather than providing solutions immediately whenever an error occurred, interns were encouraged to understand the problem, identify the cause, and find an appropriate solution.

4.2 Training Methodology

The primary methodology followed during the internship was a combination of theoretical explanation and practical implementation.
A typical training session followed the following pattern:
Concept Explanation → Practical Demonstration → Student Implementation → Assignment → Weekly Monitoring → Doubt Solving and Correction

4.2.1 Concept Explanation

The trainer first explained the concept being covered during the session. The explanation included the purpose of the concept, its basic working, and situations where it could be used.
Students maintained a notebook for important theoretical concepts, definitions, explanations, and points that were useful for future reference.

4.2.2 Practical Demonstration

After explaining a concept, the trainer demonstrated its practical implementation. This helped students understand how theoretical concepts are converted into working code or analytical outputs.
For programming topics, the demonstration was generally performed using PyCharm, while Data Science and EDA activities were primarily performed using Google Colab.

4.2.3 Student Implementation

After the demonstration, interns implemented the concepts independently on their own laptops.

This was an important part of the methodology because students were expected to write and execute the code themselves rather than simply observe the trainer's implementation.

4.2.4 Assignments

Assignments were provided based on the topics covered during the training.

The assignments were designed to encourage students to apply the concepts without continuously depending on the trainer. This helped identify areas where students required additional explanation or practice.

4.3 Weekly Assignment and Monitoring Process

The internship followed a weekly assignment-based monitoring system.
At the end of each week, students received an assignment based on the syllabus and concepts covered during that particular week.
The general process was:
Weekly Training → Practical Practice → Weekly Assignment → Submission/Implementation → Trainer Monitoring → Feedback/Correction
Assignments were not treated as formal examinations. Instead, they were used to monitor practical understanding and progress.
The trainer reviewed the work and identified areas where students required improvement. If mistakes were found, students were encouraged to understand the mistake and make the necessary corrections.
This approach helped ensure that students were keeping pace with the training rather than postponing their practical work until the end of the internship.

4.4 Python Learning Methodology

The internship began with Python programming, which provided the foundation for the subsequent Data Science activities.
The Python learning process involved:

1. Introduction to Python concepts
2. Explanation of syntax and programming logic
3. Practical coding demonstrations
4. Independent coding by students
5. Small programming problems
6. Assignment-based practice
7. Error identification and correction
8. Gradual progression toward data-oriented programming

Students initially used PyCharm IDE for basic Python programming sessions.
The focus was not only on learning Python syntax but also on developing programming logic and problem-solving ability.
Students were encouraged to write programs independently and understand why a particular approach was being used.

4.5 Data Science and EDA Methodology

After developing the necessary Python foundation, students progressed toward Data Science and Exploratory Data Analysis.

Google Colab was primarily used for this part of the internship because it provided a convenient notebook-based environment for data analysis.
The general EDA methodology followed by students was:
Dataset Selection → Data Loading → Data Understanding → Data Cleaning → Data Preprocessing → Exploratory Analysis → Data Visualization → Finding Patterns and Insights → Project Presentation

Dataset Selection

Students worked with suitable datasets that could be analyzed using Python. The datasets were generally available in CSV or Excel format.

Data Loading

The selected dataset was imported into the Python environment using appropriate tools and libraries.
Students learned how to inspect the data and understand its basic structure.

Data Understanding

Students examined the dataset to understand:

- Number of rows and columns
- Column names
- Data types
- Missing values
- Duplicate records
- Numerical and categorical data
- General characteristics of the dataset

Data Cleaning

Students performed necessary cleaning operations before proceeding with analysis.
The purpose was to ensure that the dataset was in a suitable condition for further analysis.

Data Preprocessing

Where required, students performed preprocessing operations such as handling numerical data and applying scaling techniques.
This stage helped students understand that data often requires preparation before meaningful analysis can be performed.

Exploratory Analysis

Students explored the dataset to identify patterns, relationships, distributions, and trends.
Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used during this stage.

Data Visualization

Visualizations were created to represent important characteristics of the dataset.
Students learned that visualization can make patterns easier to identify and can help communicate findings more effectively.

Finding Insights

The final objective of the EDA process was not simply to create charts but to understand what the data was indicating.
Students were encouraged to interpret their analysis and identify meaningful observations from the dataset.

4.6 Power BI Project Methodology

The second major project focused on Power BI Multipage Report Development and Analysis.

The Power BI workflow followed a structured process:
Data Collection → Data Import → Data Understanding → Data Preparation → Report Development → Visualization → Interactivity → Analysis → Final Presentation

Data Import

Students imported data primarily from CSV and Excel files into Power BI Desktop.

Data Understanding and Preparation

Students examined the available fields and prepared the data as required before creating the report.
The purpose of this stage was to ensure that the data was suitable for visualization and analysis.

Report Development

Students then began developing the Power BI report.
The project required students to create a multipage report, where different report pages represented different aspects of the data.

Visualization

Students selected suitable visualizations according to the information they wanted to communicate.
The focus was on creating reports that were not only technically correct but also easy to understand and visually organized.

Interactive Features

The Power BI project also included interactive reporting features such as:

- Slicers
- Bookmarks
- Drillthrough
- Filters and report interactions
- Page navigation and related interactive elements

These features helped students understand how a Power BI report can be designed as an interactive analytical tool rather than simply a collection of charts.

Bookmarks

Bookmarks were used to save specific report states and create interactive report experiences. Students learned how bookmarks can be used to control the visibility or state of report elements and improve report navigation.

Drillthrough

Drillthrough functionality was introduced to allow users to move from a summarized report view to a more detailed view of a particular data point or category.

This helped students understand how analytical reports can provide information at different levels of detail.

4.7 Individual Project Methodology

Both major projects were completed as individual projects.

This approach provided each intern with an opportunity to independently apply the concepts learned during the training.

Working individually required students to:

- Understand the requirements
- Work with the dataset
- Implement the required techniques
- Identify and resolve errors
- Complete the analysis/report
- Prepare the final output
- Explain their work during the presentation

Individual projects also made it easier to identify the strengths and areas of improvement of each student.

4.8 Problem-Solving Methodology

One of the important objectives of the internship was to develop the ability to identify and solve technical problems independently.

When an intern encountered an error, the immediate objective was not to provide the solution directly. Instead, students were trained to investigate the problem.

The general problem-solving approach was:

Identify the Problem → Read the Error → Understand the Cause → Try a Solution → Test the Solution → Correct the Implementation

Students were encouraged to carefully read error messages and warnings instead of ignoring them.

They were also encouraged to use available resources such as documentation, online references, and appropriate AI-based learning tools when required.

The trainer provided guidance when students were unable to proceed, but the overall approach was to make students capable of identifying errors and solving them independently.

This methodology was particularly useful during Python programming and Data Science activities, where errors in syntax, data types, file paths, data processing, or library usage were common learning opportunities.

4.9 Documentation Methodology

Documentation during the internship was kept relatively practical.
Students maintained a notebook for important theoretical concepts and explanations. The notebook was primarily used for concepts that students might need to refer to later.
However, once the internship moved toward practical Python, Data Science, and Power BI work, most of the actual work was performed digitally on the students' laptops.
For example:

- Python programs were maintained digitally.
- EDA work was performed in Google Colab.
- Power BI reports were developed in Power BI Desktop.
- Project presentations were prepared using PPTX files.
- Project-related files could also be maintained using Git/GitHub.

Only selected conceptual information, such as differences between certain Power BI concepts or important theoretical points, required extensive note-taking.
This approach allowed students to spend more time on actual implementation rather than copying large amounts of theoretical material.

4.10 Project Presentation Methodology

After completing their individual projects, students were required to present and explain their project work.
Students prepared PowerPoint presentations (PPTX) containing relevant information about their projects.
The presentation process helped students explain:

- Project objective
- Dataset or data source
- Methodology followed
- Technologies/tools used
- Analysis performed
- Visualizations or report output
- Important findings
- Final results

Students also used screen recordings of their project work while presenting their implementation.
This activity helped develop technical communication and presentation skills, which are important when explaining technical work in an academic or professional environment.

4.11 Professional Development Methodology

The final part of the internship included professional grooming and career preparation.
Students were guided on how to present themselves professionally and how to prepare their technical profile for future employment opportunities.
A major component was Resume Building, where students were guided on presenting their:

- Educational background
- Technical skills
- Internship experience

- Projects
- Certifications or relevant achievements
- Other professional information

The objective was to help students prepare a resume that represents their skills and project experience in a clear and professional manner.

4.12 Overall Methodology Followed

The overall methodology of the internship can be summarized as:

Learn → Understand → Practice → Implement → Solve Problems → Complete Assignments → Receive Feedback → Improve → Develop Project → Present the Project

→ Prepare for Professional Opportunities

The methodology gradually reduced students' dependence on direct guidance and encouraged them to become more independent in their technical work.

The combination of theory, practical implementation, weekly assignments, individual projects, problem-solving, project presentations, and professional grooming provided interns with a balanced learning experience.

The main objective throughout the process was to ensure that students did not only learn the tools, but also developed the ability to apply their knowledge, understand errors, solve problems, analyze data, communicate their findings, and work independently.

Chapter 5: Testing of Software / Tools

Testing was an important part of the practical training process. Since the internship focused on Data Science, Data Analytics, Python, and Power BI, testing was mainly carried out by running the work, checking the results, and correcting errors.

5.1 Python Program Testing

Python programs developed during the training were tested by executing the programs and checking the output. Students were encouraged to identify syntax errors, logical mistakes, and unexpected results on their own.
The main objective was to ensure that the program produced the expected output.

5.2 Data Analysis Testing

During the EDA project, students checked whether the data was properly loaded and processed. Basic checks were performed for missing values, duplicate records, incorrect data types, and unexpected results.
The generated charts and analysis were also reviewed to ensure that they represented the data correctly.

5.3 Power BI Report Testing

Power BI reports were checked after development to ensure that the visualizations, filters, slicers, bookmarks, drillthroughs, and page navigation were working properly.
Students also verified whether the displayed values and charts matched the underlying data.

5.4 Project Review

Completed projects were reviewed before the final presentation. Any noticeable errors or issues were corrected by the students.
The testing process mainly focused on accuracy, proper functioning, and correctness of the final output rather than formal software-testing procedures.

Chapter 6: Safety Procedures

Since the internship involved software development, Data Science, Data Analytics, and Power BI, the safety procedures were mainly related to the proper use of computers and maintaining a safe working environment.

6.1 Electrical Safety

Students were advised to use laptops, chargers, and electrical sockets carefully. Overloading of power connections and improper handling of charging equipment was avoided.

6.2 Computer and Workplace Safety

Students were expected to maintain a clean and organized workspace and handle laptops and other equipment carefully. Proper sitting posture was encouraged during long working sessions.

6.3 Data Handling

For training and project activities, students were provided with publicly available datasets. Actual confidential or private organizational data was not shared with students.
This ensured that students could practice data analysis while keeping real organizational information protected.

6.4 Responsible Use of Online Tools

Students were guided to use online resources and AI tools such as ChatGPT and Gemini appropriately for learning and understanding concepts. They were encouraged to understand and verify the information rather than blindly using generated content.

6.5 General Safety Practices

Students were also advised to take suitable breaks during prolonged computer usage and maintain a comfortable working position. These simple practices helped maintain a safe and productive training environment.

Chapter 7: Practical Experiences

During the 12-week industrial training, interns gained practical experience by applying the concepts learned during the training to assignments and individual projects.
The major practical experiences gained by the interns were:

- Python Programming: Writing and executing Python programs and developing programming logic.
- Data Analysis: Working with datasets and performing data cleaning, preprocessing, and exploratory analysis.
- Data Visualization: Creating and interpreting visualizations using Python libraries such as Matplotlib and Seaborn.
- Power BI: Developing multipage interactive reports using Power BI Desktop.
- Interactive Reporting: Working with features such as Slicers, Bookmarks, and Drillthroughs.
- Problem-Solving: Identifying errors, understanding their causes, and attempting to solve them independently.
- Individual Project Development: Applying the concepts learned during training to complete individual projects.
- Technical Presentation: Presenting project work using PPTX presentations and screen recordings.
- Generative AI: Understanding the basic use of Gemini and ChatGPT as supporting learning tools.
- Professional Development: Developing communication, presentation, resume-building, and professional grooming skills.

The practical nature of the training helped interns gain confidence in working independently. They were encouraged to learn from errors, apply concepts practically, and explain their own work, which provided useful preparation for entering a professional environment.

Chapter 8: Detailed Report of Tasks Undertaken

The 12-week industrial training was divided into different phases, with each phase focusing on a specific set of technical and professional skills.

8.1 Week 1 to Week 4 – Python, Data Science and EDA

During the first four weeks, interns developed their Python programming foundation and gradually moved toward Data Science and Exploratory Data Analysis.

Major tasks included:

- Learning Basic Python and Advanced Python concepts.
- Practicing Python programs and problem-solving exercises.
- Working with data using Pandas and NumPy.
- Learning Matplotlib and Seaborn for data visualization.
- Understanding basic Data Science concepts.
- Loading and examining datasets.
- Performing data cleaning and preprocessing.
- Handling missing and duplicate data.
- Performing Exploratory Data Analysis (EDA).
- Creating charts and interpreting the results.
- Working with scaling and other preprocessing techniques where required.
- Completing an individual EDA project.
- Preparing the project output for final presentation.

Students primarily used PyCharm for Python programming and Google Colab for Data Science and EDA activities.

8.2 Week 5 to Week 10 – Power BI and Data Analytics

The next six weeks focused on Power BI and Data Analytics. Students gradually progressed from basic Power BI concepts to advanced report development.

The major tasks included:

- Introduction to Power BI Desktop.
- Importing data from CSV and Excel files.
- Understanding Power BI interface and basic concepts.
- Data preparation using Power Query.
- Understanding M Language at an introductory level.
- Creating basic and advanced charts and visualizations.
- Understanding and creating DAX expressions and measures.
- Working with filters, slicers, and interactive features.
- Creating and organizing multipage reports.
- Using Bookmarks for interactive report experiences.
- Using Drillthrough to provide detailed views.
- Improving report layout and presentation.
- Understanding the basic purpose of Power BI Service.

The emphasis during this phase was on applying the concepts practically and developing confidence in creating interactive analytical reports.

8.3 Week 11 – Individual Power BI Project

During Week 11, each intern worked on an individual Power BI project using a relatively large dataset obtained from Kaggle.
The project required interns to apply the concepts covered during the previous weeks.

The major tasks included:

- Selecting and understanding the assigned dataset.
- Importing and preparing the data.
- Applying Power Query transformations where required.
- Creating appropriate visualizations.
- Applying DAX for required calculations.
- Designing a multipage Power BI report.
- Adding interactive features such as Slicers, Bookmarks, and Drillthroughs.
- Analyzing the data and identifying useful insights.
- Finalizing the report.
- Preparing a PPTX presentation.
- Recording the project demonstration and presenting the completed work.

The individual nature of the project gave each intern an opportunity to demonstrate their own understanding and practical skills.

8.4 Week 12 – Professional Grooming and Career Preparation

The final week focused mainly on professional development and career preparation.

Activities included:

- Professional Grooming
- Resume Building
- Introduction to Generative AI
- Understanding the practical use of Gemini and ChatGPT
- Guidance on presenting technical skills and projects
- Improving communication and presentation skills
- Preparing students for the professional work environment

The objective of this phase was to help interns move from technical training toward professional and career readiness.

8.5 Overall Tasks Undertaken

Across the 12-week internship, the interns progressed from basic programming to practical Data Science, EDA, Power BI development, project implementation, and professional preparation.

The overall task progression was:

Python → Data Science → EDA Project → Power BI → DAX & Power Query → Multipage Reports → Individual Power BI Project → GenAI → Professional Grooming → Resume Building

This progression provided interns with an opportunity to apply their learning continuously and develop both technical and professional skills.

Chapter 9: Special / Challenging Experiences

During the internship, interns faced several challenges that helped them improve their technical skills, problem-solving ability, and confidence.

9.1 Debugging and Error Handling

Initially, students found it difficult to identify the cause of Python errors and unexpected outputs. With regular practice, they learned to read error messages, identify the problem, and try possible solutions independently.

9.2 Data Cleaning and EDA

Working with real-world datasets introduced challenges such as missing values, duplicate records, incorrect data types, and inconsistent data. Understanding how to clean the data and decide what should be done with it required practical experience.
Another challenge was interpreting the results of EDA. Students gradually understood that creating charts is not enough; the important part is understanding what the data is showing.

9.3 Power BI Concepts

Students initially required practice to understand the difference between Power Query, DAX, and M Language and their respective purposes.
Developing interactive reports using Bookmarks, Drillthroughs, and multipage layouts also required additional practice.

9.4 Working with a Large Dataset

The Week 11 individual Power BI project involved working with a relatively large Kaggle dataset. Handling a larger dataset and converting it into a meaningful report provided a useful practical experience for the interns.

9.5 Project Presentation

Presenting individual projects was another important experience. Students had to explain their methodology, analysis, visualizations, and findings. This helped improve their technical communication and confidence.

Overall, these challenges encouraged interns to become more independent, patient, and systematic while solving technical problems

Chapter 10: Conclusion

The 12-week industrial training at Soft Tech Innovations provided interns with an opportunity to apply their academic knowledge in a practical environment. The training followed a gradual approach, beginning with Python programming and progressing toward Data Science, Exploratory Data Analysis, Power BI, DAX, Power Query, M Language, and interactive report development.

The two major individual projects gave interns practical experience in data analysis using Python and developing Power BI reports using larger datasets. Working on these projects helped them understand how theoretical concepts can be applied to practical data-related problems.

The internship also helped interns develop important skills such as problem-solving, data interpretation, visualization, project development, and technical presentation. They were encouraged to identify errors and solve problems independently, which improved their confidence and ability to work without continuous assistance.

The final phase of the training focused on Generative AI, professional grooming, and resume building, helping students prepare for future employment opportunities.

Overall, the internship provided a balanced combination of technical learning, hands-on practice, project experience, and professional development. It helped interns gain practical knowledge of the Data Science and Data Analytics domain and provided them with a better understanding of the expectations of a professional working environment.

Chapter 11: References

Sr. No.	Resource	Description	Official Link
1	Python Documentation	Official Python documentation for programming concepts and language references.	https://docs.python.org/3/
2	Pandas Documentation	Reference material for data manipulation, DataFrames, data cleaning, and analysis.	https://pandas.pydata.org/docs/
3	NumPy Documentation	Reference for numerical operations and array-based data processing.	https://numpy.org/doc/
4	Matplotlib Documentation	Reference for creating charts and data visualizations using Python.	https://matplotlib.org/stable/users/index.html
5	Seaborn Documentation	Reference for statistical data visualization and exploratory data analysis.	https://seaborn.pydata.org/
6	Microsoft Power BI Documentation	Reference material for Power BI Desktop, visualizations, Power Query, DAX, Bookmarks, Drillthrough, and Power BI Service.	https://learn.microsoft.com/power-bi/
8	Kaggle	Source of publicly available datasets used for practical Data Science and Power BI project activities.	https://www.kaggle.com/
9	Git Documentation	Reference material for understanding Git and basic version control concepts.	https://git-scm.com/doc
10	Google Colab	Online environment used for Python-based Data Science and EDA activities.	https://colab.research.google.com/